

Trends in Incidence of Invasive Lobular Carcinoma of the Breast by Race: Patterns by Age, Cancer Stage, and Socioeconomic Factors in the United States, 1992-2019

Ryann M. Quinn,¹ Ana M. Bernal,² Sun Young Oh,² Jesus D. Anampa²

Abstract

This population-based study shows that the age-adjusted incidence of ILC follows similar patterns as the overall invasive breast carcinoma, with non Hispanic White women having the largest incidence, followed by non Hispanic Black women, and then Hispanic women. However, in recent years non Hispanic Black and Hispanic women had a larger increase in ILC incidence.

Purpose: There is scarce literature about the role of race/ethnicity and socioeconomic status on the incidence of invasive lobular carcinoma (ILC). We sought to assess trends in ILC incidence by race/ethnicity across age, cancer stage, and socioeconomic status subgroups in the United States from 1992 to 2019. **Methods:** This population-based cross-sectional study included data from SEER12 registries. We used the NCI's Join point Regression Program to estimate longitudinal trends in age-adjusted breast cancer incidence rates from 1992 to 2019, reported as average annual percent change (AAPC) or annual percent change (APC). We evaluated incidence trends by a combination of race/ethnicity and stage, county-level poverty, county-level metro/nonmetro status, high school education, and age.

Results: From 1992 to 2019, ILC incidence rates increased across all race/ethnicity groups, with the greatest increase in non Hispanic Black (AAPC 2.6%), followed by Hispanic (AAPC 2.2%), and non Hispanic White women (AAPC 1.6%). The stronger increase in ILC incidence among Hispanic and non Hispanic Black women was predominantly seen among those living in low poverty or metropolitan areas and those older than 70. In recent years, from 2002 to 2019, the largest increase in ILC incidence was found in Hispanic women (APC 3.3%). **Conclusion:** Although the incidence of ILC continues to be highest in non Hispanic White women, in recent years the largest increases are seen in non Hispanic Black women and Hispanic women. Further research is warranted to better understand these trends and appropriately target at-risk populations for screening.

Clinical Breast Cancer, Vol. 25, No. 4, e394–e402 © 2024 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Keywords: Breast cancer, Obular breast cancer, Incidence

Introduction

Breast cancer is the leading cause of cancer death among non Hispanic Black and Hispanic women.¹⁻³ Breast cancer incidence rates are highest in non Hispanic White women, followed by non Hispanic Black women, and lowest in Hispanic women. Breast cancer mortality rates are higher for non Hispanic Black women,

followed by non Hispanic White women, and lowest in Hispanic women.⁴

Invasive lobular carcinoma (ILC) is the most common 'special' histological subtype of invasive breast carcinoma,⁵ accounting for 10% to 15% of all breast cancers. Multifocal or multicentric as well as contralateral diseases are more frequently found among patients with ILC.⁶ Imaging evaluation of ILC is challenging. Due to lack of opacity and low rate of calcifications, the sensitivity of mammographic to detect breast cancer is lower in ILC.⁷ Other studies have reported lower ultrasound sensitivity to detect axillary nodal metastasis in ILC.⁸ Therefore, breast magnetic resonance imaging is commonly used to evaluate the extent of ILC and to identify additional foci of disease.⁷ ILC tumors are typically hormone-receptor positive, low grade, and less sensitive to chemotherapy. While this subtype generally infers a similar or possibly more favor-

¹ Division of Hematology/Oncology, UMass Memorial Medical Center, UMass Chan Medical School, Worcester, MA

² Department of Medical Oncology, Montefiore Einstein Comprehensive Cancer Center, Bronx, New York, NY

Submitted: Oct 3, 2024; Revised: Dec 17, 2024; Accepted: Dec 28, 2024; Epub: 30 December 2024

Address for correspondence: Jesus D. Anampa, MD, 1695 Eastchester Rd, 2nd floor. Bronx, NY 10467

E-mail contact: janampa@montefiore.org

able short-term prognosis than invasive ductal carcinoma, recent data suggests worse long-term outcomes in patients with invasive lobular carcinoma.^{9,10} Although current treatment paradigms do not distinguish between ILC and the more common invasive ductal carcinoma, differences in biology and clinical features have led to increased research in this subtype and its' recognition as a unique entity.¹¹⁻¹³

Trends in the incidence of ILC have not always mimicked the trends in invasive breast cancer that have been traditionally reported by US population studies.¹⁴⁻¹⁶ This may be in part due to socioeconomic factors, environmental factors, and lifestyle risk factors having different effects in the formation of lobular neoplasms than that of other breast cancer subtypes.¹⁷ For example, hormone-related risk factors are more strongly associated with ILC. When combined hormone replacement therapy became popular among postmenopausal women, the incidence of invasive lobular carcinoma increased at a greater rate than invasive ductal carcinoma,¹⁸⁻²⁰ and similarly after the decrease in hormone replacement therapy use, the incidence of invasive lobular carcinoma decreased more rapidly.²¹

Racial disparities in breast cancer incidence, treatment, and prognosis have been reported by multiple authors; moreover, socioeconomic position may have different effects on such disparities.²²⁻²⁴ Kaur et al. reported a concerning increase in invasive breast cancer incidence among non Hispanic Black and Hispanic women, particularly among those living in poverty and rural areas.²⁵ There is scarce literature about racial disparities in ILC incidence trends in the last decade. Given the different tumor biology and potential differing risk factors, we sought to assess trends in ILC incidence by race/ethnicity across age, cancer stage, and socioeconomic status subgroups in the United States from 1992 to 2019.

Materials and Methods

Study Population

Cancer incidence data from the Surveillance, Epidemiology, and End Results Program (SEER), 12 Registries from 1992 to 2019 were obtained (2021 submission). The deidentified data were determined to be exempt from informed consent by the Albert Einstein Institutional Review Board, which approved the study. We included non Hispanic Black, non Hispanic White, and Hispanic women diagnosed with ILC (International Classification of Diseases for Oncology [ICD-O], Third Edition histological codes 8520/3). Staging information was collected using SEER summary staging, and categorized as local, regional (via direct extension or regional lymph node involvement), or distant. Demographic and socioeconomic data was also collected including age at diagnosis (< 60, 60-69, ≥ 70 years), percent of each county living below the federal poverty line (< 10%, 10%-19.99%, or ≥ 20%), urban/rural county status, and the percent of individuals in the county with at least a high school education (< 10%, 10%-14%, > 14%). County urban/rural status was defined by US Department of Agriculture's rural-urban continuum codes as follows; metropolitan counties (counties in metropolitan areas, codes 1-3) and nonmetropolitan counties (urban, codes 4-7; rural, codes 8-9; unknown, codes 88 and 99).

Statistical Analysis

Using NCI SEER*Stat (version 8.4.0.1), we extracted age-adjusted incidence rates for invasive lobular carcinoma per 100,000 women, 95% confidence intervals (CIs), and standard errors for each year from 1992 to 2019 by a combination of race/ethnicity (Hispanic, non Hispanic Black, and non Hispanic White) and stage, county-level poverty, county-level metro/nonmetro status, county level-high school education, and age. Age-adjusted incidence rates were compared to the 2000 US standard population using US Census Data (Census P25-1130). Trends for poverty, and high school education were available only from 2007 to 2019.

We used the NCI's Join point Regression Program (version 4.9.1.0) to estimate longitudinal trends in age-adjusted breast cancer incidence rates from 1992 to 2019. This software uses piecewise regression to test for significant trends over time. First, it identifies the optimal number of join points by using a Monte Carlo permutation, and then, the model segments the entire time period by those join points and estimates an annual percent change (APC) for each segment, and then calculates an average annual percent change (AAPC) for the entire study period (1992-2019).²⁶ We used a log-linear model with age-adjusted incidence rate as the outcome and years as the independent variable. We first ran the analysis on the entire study population and then by a combination of race/ethnicity and stage, poverty, county metro/nonmetro status, high school education, and age. Tests for significance were 2-sided with *P* values < .05 considered statistically significant.

Results

From 1992 to 2019, 56,540 new cases of ILC were reported among non Hispanic White, non Hispanic Black, and Hispanic women. The incidence rate was highest for non Hispanic White women (13.52 per 100,000 people), with non Hispanic Black women (8.60 per 100,000 people) and Hispanic women (7.52 per 100,000 people) having a lower incidence (Table 1). The APC in the incidence rates from 1992 to 2019 are shown in Table 2 and their curves depicted in Figure 1. In non Hispanic White women, the age-adjusted incidence rate increased from 1992 to 1998 (APC 4.1%), then decreased from 1998 to 2004 (APC -3.2%), and then increased again from 2004 to 2019 (APC 2.6%). The age-adjusted incidence rate increased during the entire study period (1992-2019) for non Hispanic Black women (APC 2.6%). Among Hispanic women, the age-adjusted incidence rate increased from 1992 to 1999 (APC 3.3%), then was unchanged between 1999 and 2002 (APC -6.6%, *P* = .429), and then increased from 2002 to 2019 (APC 3.3%). From 1992 to 2019, ILC incidence rates increased across race/ethnicity groups, with the greatest AAPC increase in non Hispanic Black (2.6%), followed by Hispanic (2.2%), and then non Hispanic White women (1.6%).

Stage at Diagnosis

The incidence rate of local stage ILC was higher in non Hispanic White (8.41 per 100,000 people), than in non Hispanic Black (4.94 per 100,000 people) or Hispanic (4.35 per 100,000 people) women. However, the incidence rate of distant stage ILC was lower in Hispanic (0.47 per 100,000 people), than in non Hispanic White

Table 1 Age-Adjusted Incidence Rates of Invasive Lobular Breast Cancer by Race/Ethnicity, Socioeconomic Status, and Education from Surveillance, Epidemiology, and End Results Program (SEER) 12 Registries, 1992-2019

	Incidence Rate (95% CI) ^a		
	Non Hispanic White	Non Hispanic Black	Hispanic
Overall	13.52 (13.40-13.64)	8.60 (8.32-8.89)	7.52 (7.32-7.72)
Stage at diagnosis			
Local	8.41 (8.32-8.51)	4.94 (4.73-5.16)	4.35 (4.20-4.51)
Regional	4.23 (4.16-4.30)	2.80 (2.65-2.97)	2.58 (2.46-2.70)
Distant	0.71 (0.69-0.74)	0.72 (0.64-0.80)	0.47 (0.42-0.52)
County-level % living in poverty			
< 10%	15.99 (15.65-16.35)	11.72 (10.59-12.94)	10.69 (9.92-11.51)
10%–19.9%	14.24 (14.02-14.47)	9.95 (9.50-10.42)	7.97 (7.67-8.27)
≥ 20%	10.44 (9.41-11.60)	7.91 (5.94-10.44)	7.32 (6.25-8.53)
County-level metro/nonmetro status			
Urban	14.05 (13.91-14.19)	8.69 (8.40-8.99)	7.53 (7.33-7.75)
Rural	10.41 (10.12-10.71)	5.27 (3.97-6.87)	7.03 (6.26-7.86)
% Highschool education			
< 10%	14.94 (14.67-15.22)	10.71 (9.89-11.57)	11.40 (10.36-12.54)
10-14%	14.82 (14.50-15.16)	10.23 (9.56-10.95)	9.99 (9.42-10.58)
> 14%	13.98 (13.59-14.39)	9.59 (8.94-10.28)	7.24 (6.93-7.57)
Age groups			
< 60 years	6.07 (5.97-6.17)	3.73 (3.55-3.93)	3.42 (3.29-3.56)
60-69 years	45.59 (44.79-46.39)	28.38 (26.62-30.22)	25.55 (24.27-26.88)
≥ 70 years	55.58 (54.79-56.38)	37.00 (34.91-39.19)	30.30 (28.81-31.84)

Abbreviations: CI = confidence interval.

^a Incidence rate per 100,000 and age-adjusted to the 2000 US standard population. (19 age groups - Census P25-1130).

(0.71 per 100,000 people) or non Hispanic Black (0.72 per 100,000 people) (Table 1).

The incidence rate of local stage ILC in non Hispanic White women initially increased from 1992 to 1998 (APC 4.2%), then decreased from 1998 to 2004 (APC −4.5%), and then increased again from 2004 to 2019 (APC 3.7%), whereas it increased in non Hispanic black women from 2002 to 2019 (APC 4.6%) and in Hispanic women from 2003 to 2019 (APC 3.6%) (Table 2 and Supplemental Figure S1). The age-adjusted incidence rate of regional stage ILC in non Hispanic White increased from 1992 to 1999 (APC 4.6%), remained unchanged from 1999 to 2002 (APC −4.5%, $P = .509$), and then increased again from 2002 to 2019 (APC 0.7%). Similar trends were seen in Hispanic women (APC for 1992-2002, 5.2%; APC for 2000-2004, −4.2%, $P = .549$; APC for 2004-2019, 2.4%). Non Hispanic Black women had increase in incidence through the study period (APC 2.2%). The age-adjusted incidence rate of distant stage ILC in non Hispanic White women increased from 1992 to 2013 (APC 3.6%) and then remained unchanged from 2013 to 2019 (APC −4.4% $P = .118$). Whereas non Hispanic Black (APC 2.6%) and Hispanic (APC 2.2%) women had an increase in incidence rate throughout the entire study period.

From 1992 to 2019, all racial/ethnic groups had an increase in the incidence rate of local stage ILC (AAPC for non Hispanic White 1.9%; AAPC for non Hispanic Black 2.1%; AAPC for Hispanic 1.8%) and distant stage ILC (AAPC for non Hispanic White 1.8%; AAPC for non Hispanic Black 2.6%; AAPC for Hispanic 2.2%). Only non Hispanic Black

women had an increase in incidence in regional disease (AAPC 2.2%).

County-Level Poverty

For all county-level poverty categories, the age-adjusted incidence rate of ILC was the highest among non Hispanic White women, followed by non Hispanic Black women, and then Hispanic women (Table 1). Trends in ILC incidence rates by poverty were available from 2007 onward (Table 2 and Supplemental Figure S2). The age-adjusted incidence rate of ILC in counties with < 10% poverty increased annually by 2.3% in non Hispanic White, by 4.3% in non Hispanic Black, and by 3.6% in Hispanic women from 2007 to 2019. Similarly, the incidence rate of ILC in counties with 10% to 19.99% poverty increased for non Hispanic White (AAPC 2.3%), non Hispanic Black (AAPC 2.1%), and Hispanic women (AAPC 2.1%) from 2007 to 2019. The incidence rate of ILC in counties with ≥ 20% poverty increased in non Hispanic White (AAPC 3.3%), whereas it was unchanged for non Hispanic Black (AAPC −0.7% $P = .729$) and Hispanic women (AAPC 4% $P = .074$). It is noteworthy that the number of cases was small for all race/ethnicity groups in counties with > 20% poverty, especially non Hispanic Black and Hispanic women. The trends in ILC incidence rate across race/ethnicity and poverty subgroups were driven primarily by incidence trends in local and regional disease with nonsignificant contribution from the distant stage ILC (Figure 2 and Supplemental Table S1).

Table 2 Trends in Invasive Lobular Breast Cancer Incidence Rates by Race/Ethnicity Stratified by Stage, Socioeconomic Status and Education from Surveillance, Epidemiology, and End Results Program (SEER) 12 Registries, 1992-2019

Strata	Race/Ethnicity	No.	APC						AAPC	
			Trend 1		Trend 2		Trend 3		1992-2019	2007-2019
			Years	APC	Years	APC	Years	APC		
Overall	Non Hispanic White	47,209	1992-1998	+4.1*	1998-2004	-3.2*	2004-2019	+2.6*	+1.6*	
	Non Hispanic Black	3,645	1992-2019	+2.6*					+2.6*	
	Hispanic	5,686	1992-1999	+3.3*	1999-2002	-6.6	2002-2019	+3.3*	+2.2*	
Stage										
Local	Non Hispanic White	29,598	1992-1998	+4.2*	1998-2004	-4.5*	2004-2019	+3.7*	+1.9*	
	Non Hispanic Black	2,072	1992-2002	-1.9	2002-2019	+4.6*			+2.1*	
	Hispanic	3,247	1992-2003	-0.8	2003-2019	+3.6*			+1.8*	
Regional	Non Hispanic White	14,501	1992-1999	+4.6*	1999-2002	-4.5	2002-2019	+0.7*	1.1	
	Non Hispanic Black	1,207	1992-2019	+2.2*					+2.2*	
	Hispanic	1,994	1992-2000	+5.2*	2000-2004	-4.2	2004-2019	+2.4*	2.2	
Distant	Non Hispanic White	2,513	1992-2013	+3.6*	2013-2019	-4.4			+1.8*	
	Non Hispanic Black	311	1992-2019	+2.6*					+2.6*	
	Hispanic	361	1992-2019	+2.2*					+2.2*	
County-level% living in poverty ^a										
< 10%	Non Hispanic White	8,611	2007-2019	+2.3*						+2.3*
	Non Hispanic Black	417	2007-2019	+4.3*						+4.3*
	Hispanic	776	2007-2019	+3.6*						+3.6*
10%–19.9%	Non Hispanic White	16,385	2007-2019	+2.3*						+2.3*
	Non Hispanic Black	1,937	2007-2019	+2.1*						+2.1*
	Hispanic	2,867	2007-2019	+2.1*						+2.1*
≥20%	Non Hispanic White	417	2007-2019	+3.3*						+3.3*
	Non Hispanic Black	58	2007-2019	-0.7						-0.7
	Hispanic	171	2007-2019	4						4
County-level metropolitan status										
Metro	Non Hispanic White	41,795	1992-1998	+4.0*	1998-2004	-3.5*	2004-2019	+2.6*	+1.5*	
	Non Hispanic Black	3,587	1992-2019	+2.7*					+2.7*	
	Hispanic	5,354	1992-2004	-0.1	2004-2019	+3.1*			+1.7*	
Nonmetro	Non Hispanic White	5,280	1992-2019	+1.2*					+1.2*	
	Non Hispanic Black	56	1992-2019	-0.6					-0.6	
	Hispanic	313	1992-2019	+1.6*					+1.6*	
% Highschool education ^a										
< 10%	Non Hispanic White	11,873	2007-2019	+2.4*						+2.4*
	Non Hispanic Black	684	2007-2019	+4.7*						+4.7*
	Hispanic	502	2007-2019	1.4						1.4
10-14%	Non Hispanic White	8,324	2007-2011	+4.9*	2011-2019	+2.1*			+3.0*	
	Non Hispanic Black	903	2007-2019	1.8					1.8	
	Hispanic	1,229	2007-2019	+2.5*					+2.5*	
> 14%	Non Hispanic White	5,216	2007-2019	+2.2*					+2.2*	
	Non Hispanic Black	825	2007-2019	1.6					1.6	
	Hispanic	2,083	2007-2019	+2.7*					+2.7*	
Age groups										
< 60 years	Non Hispanic White	15,417	1992-1998	+3.8*	1998-2003	-3.4	2003-2019	+2.2*	+1.5*	

(continued on next page)

Trends in Incidence of Invasive Lobular Carcinoma

Table 2 (continued)

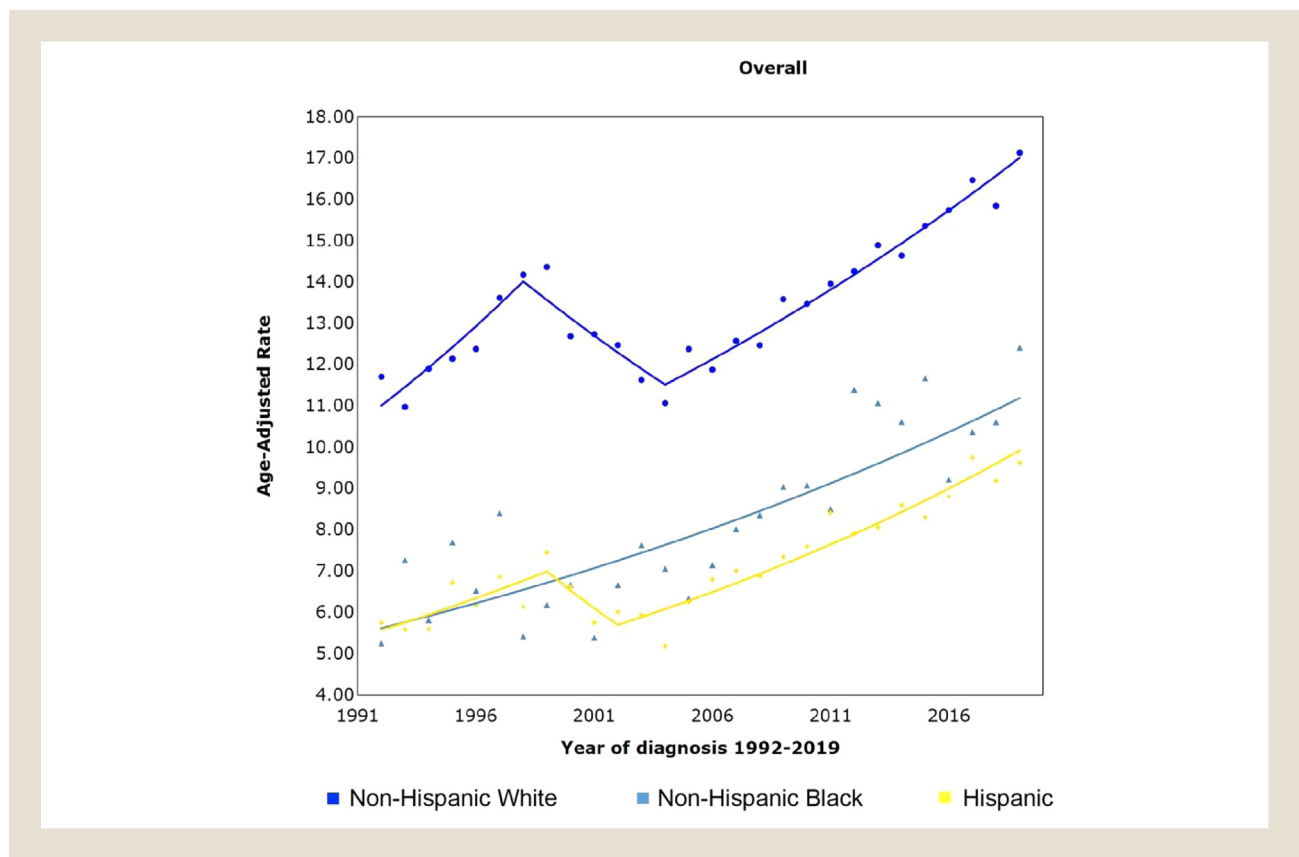
Strata	Race/Ethnicity	No.	APC						AAPC	
			Trend 1		Trend 2		Trend 3		1992-2019	2007-2019
			Years	APC	Years	APC	Years	APC		
60-69 years	Non Hispanic Black	1,498	1992-2019	+2.0*					+2.0*	
	Hispanic	2,630	1992-1999	+6.1*	1999-2002	−8.2	2002-2019	+3.2*	2.6	
	Non Hispanic White	12,623	1992-1999	+4.4*					1.7	
	Non Hispanic Black	979	1992-2019	+2.8*	1999-2003	−5.9	2003-2019	+2.4*	+2.8*	
≥ 70 years	Hispanic	1,495	1992-2019	+2.0*					+2.0*	
	Non Hispanic White	19,169	1992-1999	+3.4*	1999-2004	−4.7*	2004-2019	+3.1*	+1.7*	
	Non Hispanic Black	1,168	1992-2019	+3.1*					+3.1*	
	Hispanic	1,561	1992-2019	+1.9*					+1.9*	

Abbreviations: APC = annual percent change; AAPC = average annual percent change.

* Statistically significant ($P < .05$).

^a Trends for county-level poverty, household income, and high school education were available only from 2007 to 2019.

Figure 1 Age-adjusted overall incidence rates of invasive lobular breast cancer by race/ethnicity. Age-adjusted incidence rate per 100,000 individuals.

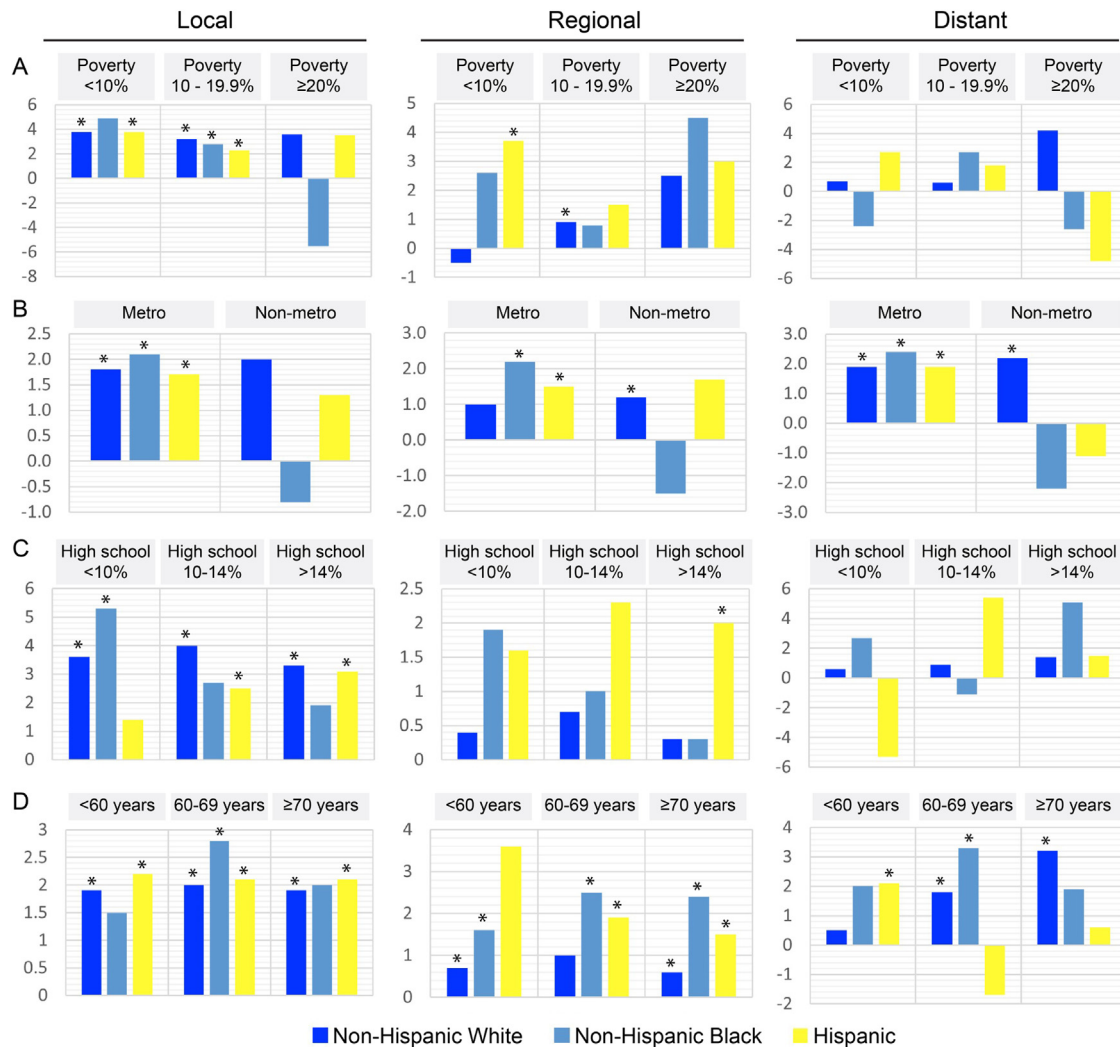


County Metro/Nonmetro Status

The age-adjusted incidence of ILC in metro counties was the highest for non Hispanic White women, followed by non Hispanic Black women, and then Hispanic women, whereas the incidence in nonmetro counties was the highest in non Hispanic White women, followed by Hispanic, and then non Hispanic Black women (Table 1).

In metro counties, the age-adjusted incidence rate of ILC in non Hispanic White women increased from 1992 to 1998 (APC 4.0%), then decreased from 1998 to 2004 (APC −3.5%), and then increased from 2004 to 2019 (APC 2.6%) (Table 2 and Supplemental Figure S3). In non Hispanic Black women, the ILC incidence rate increased from 1992 to 2019 (APC 2.7%), whereas it only increased from 2004 to 2019 (APC 3.1%) in Hispanic

Figure 2 Average annual percent changes in the age-adjusted incidence rate of invasive lobular carcinoma by race/ethnicity and stage stratified by socioeconomic factors and age. Average annual percent changes in the age-adjusted incidence rate of invasive lobular carcinoma from 1992 to 2019 by race/ethnicity and stage stratified by (A) poverty, (B) metro/nonmetro status, (C) high school education, and (D) age from Surveillance, Epidemiology, and End Results Program (SEER) 12 Registries, 1992–2019 are shown. Trends for county-level poverty were available only from 2007 to 2017. *Statistically significant ($P < .05$).



women. Overall, during the entire study period of 1992 to 2019, the incidence of ILC increased in all race/ethnic groups, particularly non Hispanic Black women (AAPC for non Hispanic White 1.5%; AAPC for non Hispanic Black 2.7%; AAPC for Hispanic 1.7%).

In nonmetro counties, the age-adjusted incidence rate of ILC increased for non Hispanic White (AAPC 1.2%) and Hispanic women (AAPC 1.6%). There was no significant trend in non Hispanic Black women, likely due to the small number of women in this group.

In metro counties, the ILC incidence trends showed a larger increase for non Hispanic Black women in local, regional, and distant stage as compared to non Hispanic White or Hispanic

women. In nonmetro counties, the trends were unchanged for all race/ethnic groups in local stage ILC; however, the ILC trends evidenced a larger increase for non Hispanic White in regional and distant stage ILC as compared to their non Hispanic Black and Hispanic counterparts (Figure 2 and Supplemental Table S1).

High School Education

Across all the subgroups of high school education, the age-adjusted incidence of ILC was the highest for non Hispanic white as compared to non Hispanic Black and Hispanic women (Table 1). In counties with > 14% high school education, the trends in incidence showed a large increase in Hispanic women (AAPC 2.7%) as

Trends in Incidence of Invasive Lobular Carcinoma

compared to non Hispanic Black (AAPC 1.6%) and non Hispanic White women (AAPC 2.2%) (Table 2 and Supplemental Figure S4). In counties with < 10% high school education, the ILC trends showed a larger increase in non Hispanic Black women (AAPC 4.7%) as compared to non Hispanic White (AAPC 2.4%) and Hispanic (AAPC 1.4%, $P = .304$) women.

In local stage ILC, non Hispanic Black women had the largest increase in ILC in counties with < 10% high school education (AAPC, 5.3%), whereas non Hispanic White women had the largest increase in counties with 10 to 14% (AAPC 4.0%) and > 14% high school education (AAPC 3.3%). In distant stage ILC, the incidence rates were unchanged across all race/ethnic and high school groups (Figure 2 and Supplemental Table S1).

Age

Across all age subgroups, the age-adjusted incidence of ILC was the highest for non Hispanic White women, followed by non Hispanic Black women, and then Hispanic women (Table 1). In women < 60 years, the age-adjusted incidence rate of ILC increased in non Hispanic white (AAPC 1.5%) and non Hispanic Black (AAPC 2.0%), whereas it remained unchanged in Hispanic women (AAPC 2.6% $P = .087$). In women 60 to 69 years, the incidence of ILC increased in non Hispanic Black (AAPC 2.8%) and Hispanic women (AAPC 2.0%), and it remained unchanged in non Hispanic White (AAPC 1.7% $P = .105$). In women ≥ 70 years, the incidence of ILC increased for all race/ethnic subgroups, with the largest increase in non Hispanic Black women (AAPC 3.1%) (Table 2 and Supplemental Figure S5).

In local stage ILC, Hispanic women had the largest increase in ILC incidence rates in women < 60 years (AAPC 2.2%) and ≥ 70 years (AAPC 2.1%), whereas non Hispanic Black had the largest increase in women 60 to 69 years (AAPC 2.8%). In regional stage ILC, non Hispanic Black had the largest increase in ILC incidence across all age subgroups (AAPC 1.6% in women < 60, AAPC 2.5% for women 60 to 69, and 2.4% for women ≥ 70 years). In distant stage ILC, Hispanic women had the greatest increase in women < 60 years (AAPC 2.1%), non Hispanic Black women had the greatest increase in women 60 to 69 years (AAPC 3.3%), and non Hispanic White women had the greatest increase in women ≥ 70 years (AAPC 3.2%) (Figure 2 and Supplemental Table S1).

Discussion

Our study reports trends in incidence of ILC of the breast by race; evaluating patterns by age, cancer stage, and socioeconomic factors in the United States from 1992 to 2019. Our study reports that the age-adjusted incidence of ILC follows similar patterns as the overall invasive breast carcinoma,⁴ with non Hispanic White women having the largest incidence, followed by non Hispanic Black women, and then Hispanic women. From 1992 to 2019, the incidence of ILC increased in all race/ethnic groups, with the largest increase found in non Hispanic Black women, followed by Hispanic women. The stronger increases among non Hispanic Black women and Hispanic women were particularly among those living in low poverty areas, metropolitan areas, and in women older than 70 years.

Recent studies have reported that the incidence of invasive breast cancer has increased in local (0.9% per year) and distant stage (0.9%

per year) but decreased in regional stage (-0.7% per year) from 2015 to 2019.⁴ Furthermore, Kaur et al. reported similar trends in incidence of invasive breast cancer by cancer stage across race/ethnic subgroups.²⁵ The change in incidence in local and regional disease is attributed to earlier detection by screening mammography,^{27,28} and the increase in distant disease is hypothesized to be due to improvement in stage classification since the rate of unknown stage has declined in parallel; however, a study of young women suggests that some of the increase in distant stage disease was real, especially among non Hispanic Black women.²⁹ In our study, we found that the incidence of ILC across all race/ethnic subgroups has increased in local, regional, and distant stage from 2004 to 2019. Moreover, larger incidence increases occurred in non Hispanic Black and Hispanic than in non Hispanic White across all stages. ILC tends to have low density of tumor cells and lack desmoplastic stromal reaction which accounts for the difficulty in its detection on physical exam and mammography.^{30,31} The sensitivity of mammography for the detection of ILC is only 57% to 79% compared to 63% to 98% for all invasive breast cancer.³² These differences may explain why the incidence of regional ILC has not decreased as seen in invasive breast cancer. Moreover, dense breast tissue may have a stronger negative effect on detecting ILC than invasive ductal carcinoma (sensitivity of mammogram for invasive ductal carcinoma in dense or heterogeneously dense breasts, 60%-86%; sensitivity for ILC, 11%-36%).³³ The stronger increase in ILC incidence for non Hispanic Black and Hispanic may be explained by the effectiveness of mammography screening in these groups due to differences in breast tissue and other biological factors.³⁴

Prior studies have reported decreased incidence in invasive breast cancer from 2001 to 2004 that was largely attributed to the decreased use of menopausal hormones after publication of results from the Women's Health Initiative randomized trial linking menopausal hormone use to breast cancer.³⁵ Kaur et al. reported that around that period of time, the incidence of invasive breast cancer decreased in non Hispanic White and Hispanic women, with no change in non Hispanic Black women. In our study, we found that the incidence of ILC decreased from 1998 to 2004 only for non Hispanic white women, while it increased in non Hispanic Black and remained stable in Hispanic women. Our results show that a decrease in menopausal hormones had similar effects on ILC incidence trends as on invasive breast carcinoma where the decrease in incidence from 2001 to 2004 was seen primarily in non Hispanic White women.³⁶

Furthermore, in the recent years, from 2002 to 2019, the largest increase in ILC incidence was seen in Hispanic women (APC, 3.3%) as compared to non Hispanic Black (APC, 2.6%) and non Hispanic White (APC, 2.6%) women. On the other hand, Kaur et al. reported that the incidence of invasive carcinoma of the breast had larger increase in non Hispanic Black women in the recent years from 2004 to 2017 (APC for non Hispanic Black, 0.9%; APC for non Hispanic White, 0.4%; APC for Hispanics, 0.6%). The subgroups driving the larger increase in ILC incidence in Hispanic women include, regional stage disease, nonmetro county, > 14% high school education, and age < 60 years. These findings are concerning given that the Hispanic population is growing fast in the US. The increase in incidence of ILC is consistent with prior

reports^{15,19,21}; however, these prior studies evaluated ILC incidence trends in early 2000s and have not examined these trends within different socioeconomic and ethnic groups. It is suspected that the rise in breast cancer recently is due to increases in hormone related risk factors, such as decreased fertility rates, increased age at first birth, and obesity.³⁷ Our study suggests that across all ethnicity/race subgroups, ILC had a larger increase in incidence in recent years when compared to invasive breast cancer incidence trends reported by Kaur et al. The higher increase in ILC compared to invasive breast cancer could be due to the fact that hormone related risk factors tend to be more associated to ILC than invasive ductal carcinoma.^{38,39} In addition, alcohol use may be more strongly associated with ILC than invasive ductal carcinoma.^{40,41} This trend may become more significant given the recent increased use of alcohol among women.⁴² Over the past 50 years, studies have reported an increase in obesity rates in the US.⁴³ Obesity is another known risk factor for breast cancer, though the association between obesity and breast cancer is less clear in premenopausal women. Due to lack of information in our database, we were unable to include body mass index in our analysis; therefore, we were unable to isolate the effect of obesity on our results. A recent study showed that obesity prevalence was inversely associated with median household income in White and Black populations at the population level of large US cities,⁴⁴ which highlights that the interplay of socioeconomic factors in the association of race and ILC can be complex and multifactorial. Multiple studies have shown an inverse association between physical activity and breast cancer risk.⁴⁵⁻⁴⁸ Physical activity measurement was not available in our data set but further investigation between ILC incidence and physical activity among different racial backgrounds would provide more insight in racial disparities in breast cancer incidence and outcome. Another potential factor in ILC incidence rate differences among racial groups is hormonal uptake, which prior studies have demonstrated to have a strong association with ILC; however, this information was not available in our database. Our study results highlight that the incidence of ILC is increasing, but racial disparities in ILC exist and may be affected by socioeconomic factors; furthermore, the increase in the incidence of regional ILC warrants further studies to assess strategies to detect ILC earlier.

Our study has certain limitations. First, we used the incidence - SEER research database, 12 registries which covers only 12.2% of the US population; however, this database provides longitudinal data from 1992 to 2019 which allowed us to assess inflection points in the incidence trends over 3 decades. Second, the socioeconomic variables in our study were collected from cancer registries which only include neighborhood factors (county-level data); therefore, we were unable to assess the effect of individual socioeconomic factors on ILC incidence trends. Third, the number of cases of ILC stratified by race/ethnic was small in some of the socioeconomic subgroups and could have affected the results of those analyses. Fourth, we limited our analysis to non Hispanic White, non Hispanic Black, and Hispanic women; other ethnic groups were not studied. Fifth, a histologically homogeneous population was included to decrease confounding, and our analysis was restricted to classic ILC; therefore, our results cannot be extrapolated to other ILC variants such as pleomorphic ILC.

Conclusions

The incidence of ILC continues to be highest in non Hispanic White women. However, the incidence has increased across all race/ethnic groups, with the largest increases found in non Hispanic Black women, followed by Hispanic women, especially those in low poverty areas, metropolitan counties, and age ≥ 70 years. In recent years from 2002 to 2019, the largest increase in ILC incidence was found in Hispanic women. The increase in the incidence of regional disease for ILC in recent years is concerning given that the incidence of regional disease has decreased for invasive ductal carcinoma. These ILC incidence trends warrant further research to determine potential causative factors and to appropriately target groups for screening.

Clinical Practice Points

- Multiple studies have reported racial/ethnic disparities in breast cancer incidence and survival outcomes.
- Most studies that evaluate incidence trends of breast cancer by socioeconomic factors include all histological types. However, studies have shown differences in tumor biology, clinical presentation, and survival outcomes between invasive lobular carcinoma (ILC) and invasive ductal carcinoma.
- We report recent trends in ILC incidence by race/ethnicity across age, cancer stage, and socioeconomic status subgroups in the United States from 1992 to 2019.
- The incidence of ILC continues to be highest in non Hispanic White women. However, the incidence has increased across all race/ethnic groups, with the largest increases found in non Hispanic Black women, followed by Hispanic women, especially those in low poverty areas, metropolitan counties, and age ≥ 70 years.

Disclosure

All authors declare no conflict of interest during the conduction of this study.

CRediT authorship contribution statement

Ryann M. Quinn: Writing – original draft, Project administration, Conceptualization. **Ana M. Bernal:** Writing – review & editing, Project administration, Data curation, Conceptualization. **Sun Young Oh:** Writing – original draft, Supervision, Methodology, Conceptualization. **Jesus D. Anampa:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Data Availability

This study was conducted using matrices obtained from SEER*Stat 8.4.0.1. We collected the data from the Cancer incidence data from the Surveillance, Epidemiology, and End Results Program (SEER), 12 Registries from 1992 to 2019 (2021 submission). The spreadsheet created for the data analysis is available for the public after a request for SEER software from the NCI is granted. The

Trends in Incidence of Invasive Lobular Carcinoma

selection criteria necessary to obtain the matrix we used for this study are specified in the methods section of the manuscript.

Ethics Approval

This study was approved by the Institutional Board Review (IRB) of the Albert Einstein College of Medicine. Due to the retrospective nature of this study, no informed consent was required.

Permission to Reproduce Material From Other Sources

The authors have created all materials in this manuscript (tables and figures).

Clinical Trial Registration

Not applicable.

Acknowledgments

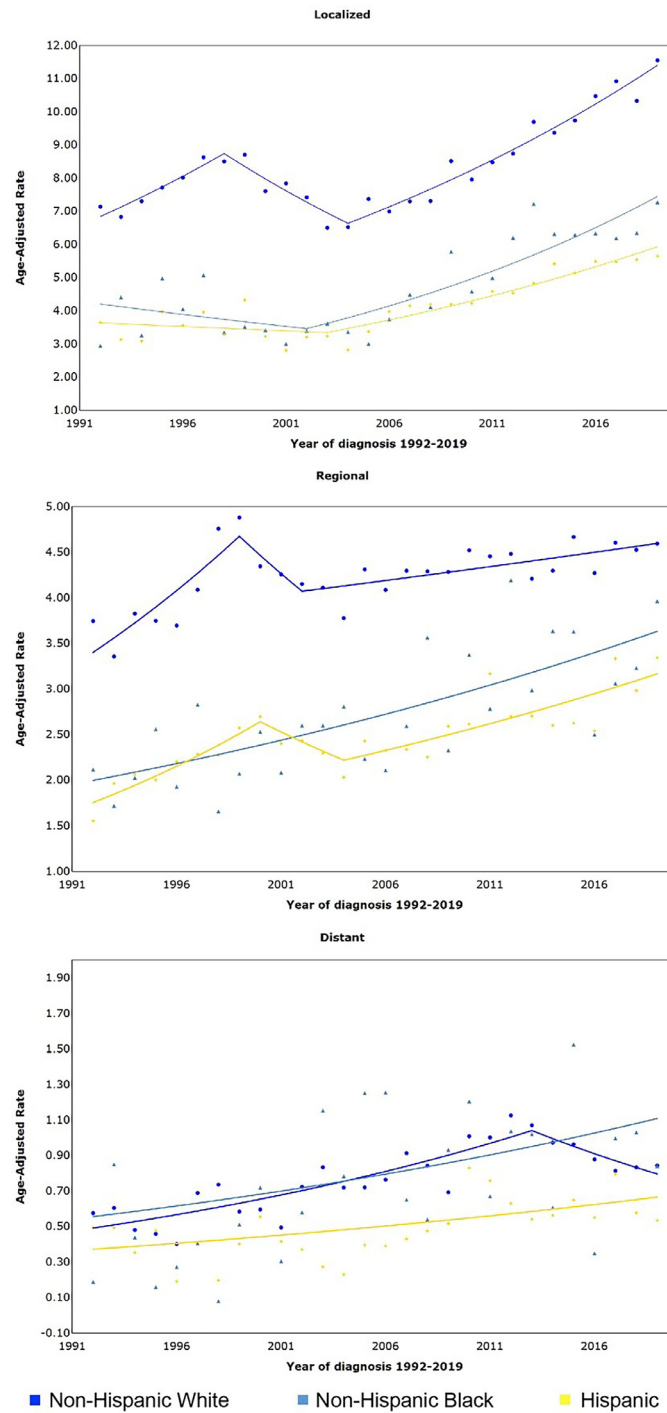
The project described was supported by the Einstein Paul Calabresi Career Development Program (NIH [5K12CA132783-08](#)) and the National Center for Advancing Translational Sciences (NCATS), National Institutes of Health, through CTSA award number (UL1 [TR002556](#)).

References

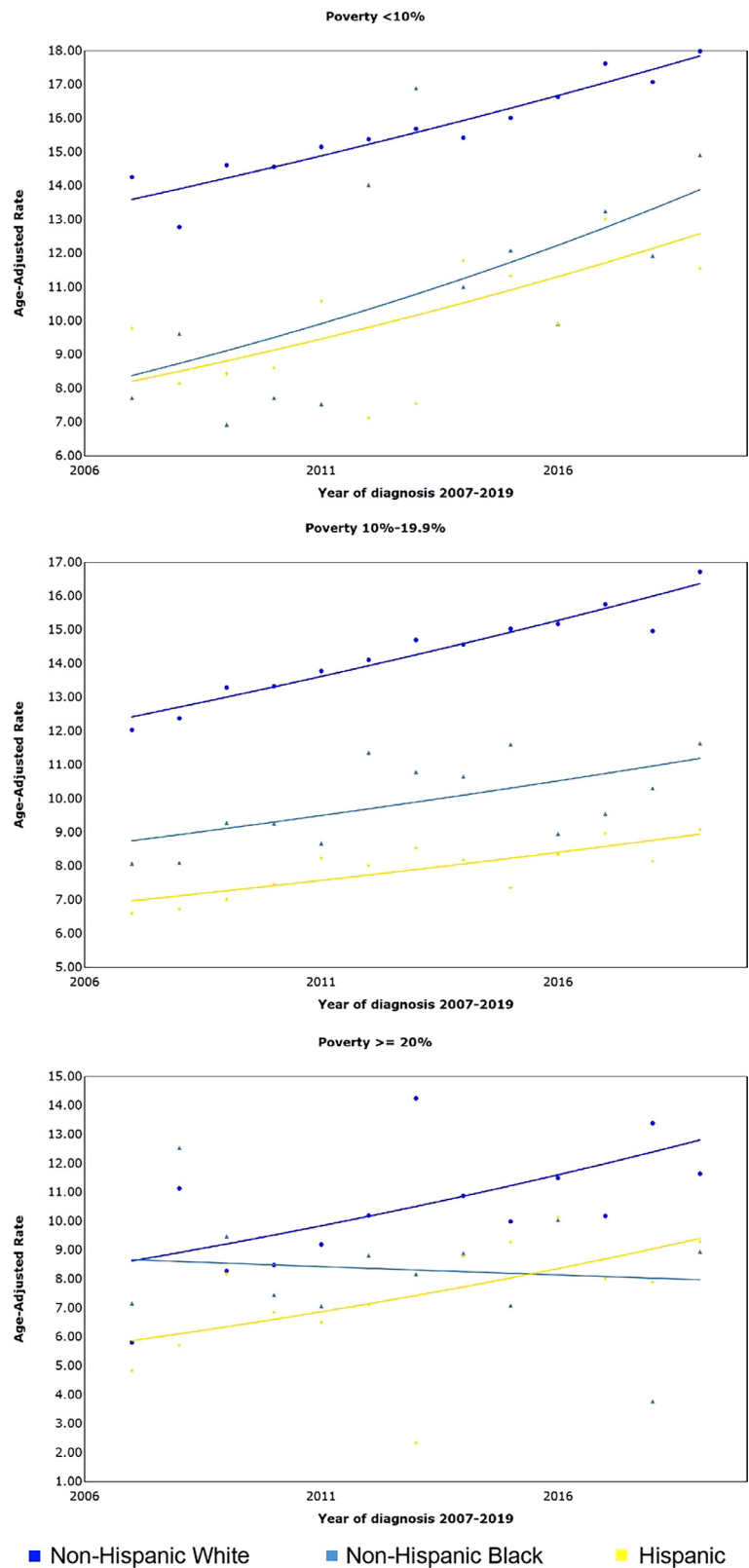
- Giaquinto AN, Miller KD, Tossas KY, Winn RA, Jemal A, Siegel RL. Cancer statistics for African American/Black People 2022. *CA: Cancer J Clin.* 2022;72:202–229.
- Miller KD, Ortiz AP, Pinheiro PS, et al. Cancer statistics for the US Hispanic/Latino population, 2021. *CA: Cancer J Clin.* 2021;71:466–487.
- Siegel RL, Miller KD, Wagle NS, Jemal A. Cancer statistics, 2023. *CA: Cancer J Clin.* 2023;73:17–48.
- Giaquinto AN, Sung H, Miller KD, et al. Breast Cancer Statistics, 2022. *CA: Cancer J Clin.* 2022;72:524–541.
- Cserni G. Histological type and typing of breast carcinomas and the WHO classification changes over time. *Pathologica.* 2020;112:25–41.
- Dedes KJ, Fink D. Clinical presentation and surgical management of invasive lobular carcinoma of the breast. *Breast Disease.* 2008;30:31–37.
- Cocco D, ElSherif A, Wright MD, et al. Invasive Lobular Breast Cancer: data to Support Surgical Decision Making. *Ann Surg Oncol.* 2021;28:5723–5729.
- Upadhyaya VS, Lim GH, Chan EYK, Fook-Chong SMC, Leong LCH. Evaluating the preoperative breast cancer characteristics affecting the accuracy of axillary ultrasound staging. *Breast J.* 2020;26:162–167.
- Pramod N, Nigam A, Basree M, et al. Comprehensive Review of Molecular Mechanisms and Clinical Features of Invasive Lobular Cancer. *Oncologist.* 2021;26:e943–e953.
- Luo Y, Ma A, Huang S, Yu Y. Invasive Lobular Carcinoma Has Worse Outcome Compared with Invasive Ductal Carcinoma in Stage IV Breast Cancer with Bone-Only Metastasis. *Breast Care.* 2022;17:296–305.
- Lamovec J, Bracko M. Metastatic pattern of infiltrating lobular carcinoma of the breast: an autopsy study. *J Surg Oncol.* 1991;48:28–33.
- McCart Reed AE, Kutasovic JR, Lakhani SR, Simpson PT. Invasive lobular carcinoma of the breast: morphology, biomarkers and 'omics. *Breast cancer research : BCR.* 2015;17:12.
- McCart Reed AE, Kalinowski L, Simpson PT, Lakhani SR. Invasive lobular carcinoma of the breast: the increasing importance of this special subtype. *Breast Cancer Res.* 2021;23:6.
- Findlay-Shirras LJ, Lima I, Smith G, Clemons M, Arnaout A. Population trends in Lobular Carcinoma of the Breast: the Ontario Experience. *Ann Surg Oncol.* 2020;27:4711–4719.
- Li CI, Anderson BO, Porter P, Holt SK, Daling JR, Moe RE. Changing incidence rate of invasive lobular breast carcinoma among older women. *Cancer.* 2000;88:2561–2569.
- Verkooijen HM, Fioretta G, Vlastos G, et al. Important increase of invasive lobular breast cancer incidence in Geneva, Switzerland. *Int J Cancer.* 2003;104:778–781.
- Dossus L, Benusioglio PR. Lobular breast cancer: incidence and genetic and non-genetic risk factors. *Breast Cancer Res.* 2015;17:37.
- Hentschel S, Heinz J, Schmid-Hopfner S, et al. The impact of menopausal hormone therapy on the incidence of different breast cancer types—data from the Cancer Registry Hamburg 1991–2006. *Cancer Epidemiol.* 2010;34:639–643.
- Li CI, Anderson BO, Daling JR, Moe RE. Trends in incidence rates of invasive lobular and ductal breast carcinoma. *Jama.* 2003;289:1421–1424.
- Reeves GK, Beral V, Green J, Gathani T, Bull D. Million Women Study C. Hormonal therapy for menopause and breast-cancer risk by histological type: a cohort study and meta-analysis. *Lancet Oncol.* 2006;7:910–918.
- Eheman CR, Shaw KM, Ryerson AB, Miller JW, Ajani UA, White MC. The changing incidence of in situ and invasive ductal and lobular breast carcinomas: United States, 1999–2004. *Cancer Epidemiol Biomarkers Prev.* 2009;18:1763–1769.
- Alvarez A, Bernal AM, Anampa J. Racial disparities in overall survival after the introduction of cyclin-dependent kinase 4/6 inhibitors for patients with hormone receptor-positive, HER2-negative metastatic breast cancer. *Breast Cancer Res Treat.* 2023;198:75–88.
- Bhargava A, Du XL. Racial and socioeconomic disparities in adjuvant chemotherapy for older women with lymph node-positive, operable breast cancer. *Cancer.* 2009;115:2999–3008.
- Hershman D, McBride R, Jacobson JS, et al. Racial disparities in treatment and survival among women with early-stage breast cancer. *J Clin Oncol.* 2005;23:6639–6646.
- Kaur M, Joshi CE, Visvanathan K, Connor AE. Trends in breast cancer incidence rates by race/ethnicity: Patterns by stage, socioeconomic position, and geography in the United States, 1999–2017. *Cancer.* 2022;128:1015–1023.
- Kim HJ, Fay MP, Feuer EJ, Midthune DN. Permutation tests for jointpoint regression with applications to cancer rates. *Stat Med.* 2000;19:335–351.
- Bleyer A, Welch HG. Effect of three decades of screening mammography on breast-cancer incidence. *N Engl J Med.* 2012;367:1998–2005.
- Welch HG, Prorok PC, O'Malley AJ, Kramer BS. Breast-Cancer Tumor Size, Overdiagnosis, and Mammography Screening Effectiveness. *N Engl J Med.* 2016;375:1438–1447.
- DeSantis CE, Ma J, Jemal A. Trends in stage at diagnosis for young breast cancer patients in the United States. *Breast Cancer Res Treat.* 2019;173:743–747.
- Dabbs DJ, Schnitt SJ, Geyer FC, et al. Lobular neoplasia of the breast revisited with emphasis on the role of E-cadherin immunohistochemistry. *Am J Surg Pathol.* 2013;37:e1–11.
- Johnson K, Sarma D, Hwang ES. Lobular breast cancer series: imaging. *Breast Cancer Res.* 2015;17:94.
- Porter AJ, Evans EB, Foxcroft LM, Simpson PT, Lakhani SR. Mammographic and ultrasound features of invasive lobular carcinoma of the breast. *J Med Imaging Radiat Oncol.* 2014;58:1–10.
- Berg WA, Gutierrez L, NessAiver MS, et al. Diagnostic accuracy of mammography, clinical examination, US, and MR imaging in preoperative assessment of breast cancer. *Radiology.* 2004;233:830–849.
- Yeaman TJ, Cantor AB, Smith TJ, et al. Tumor biology of infiltrating lobular carcinoma. Implications for management. *Ann Surg.* 1995;222:559–561.
- Ravdin PM, Cronin KA, Howlader N, et al. The decrease in breast-cancer incidence in 2003 in the United States. *N Engl J Med.* 2007;356:1670–1674.
- DeSantis C, Howlader N, Cronin KA, Jemal A. Breast cancer incidence rates in U.S. women are no longer declining. *Cancer Epidemiol Biomarkers Prev.* 2011;20:733–739.
- Pfeiffer RM, Webb-Vargas Y, Wheeler W, Gail MH. Proportion of U.S. Trends in Breast Cancer Incidence Attributable to Long-term Changes in Risk Factor Distributions. *Cancer Epidemiol Biomarkers Prev.* 2018;27:1214–1222.
- Chen CL, Weiss NS, Newcomb P, Barlow W, White E. Hormone replacement therapy in relation to breast cancer. *Jama.* 2002;287:734–741.
- Kotsopoulos J, Chen WY, Gates MA, Tworoger SS, Hankinson SE, Rosner BA. Risk factors for ductal and lobular breast cancer: results from the nurses' health study. *Breast Cancer Res.* 2010;12:R106.
- Li CI, Chlebowski RT, Freiberg M, et al. Alcohol consumption and risk of postmenopausal breast cancer by subtype: the women's health initiative observational study. *J Natl Cancer Inst.* 2010;102:1422–1431.
- Li CI, Daling JR, Malone KE, et al. Relationship between established breast cancer risk factors and risk of seven different histologic types of invasive breast cancer. *Cancer Epidemiol Biomarkers Prev.* 2006;15:946–954.
- White AM. Gender Differences in the Epidemiology of Alcohol Use and Related Harms in the United States. *Alcohol Res.* 2020;40:01.
- Kranjac AW, Kranjac D. Explaining adult obesity, severe obesity, and BMI: five decades of change. *Heliyon.* 2023;9:e16210.
- Benusi M, Cheskin LJ. Obesity prevalence in large US cities: association with socioeconomic indicators, race/ethnicity and physical activity. *J Public Health (Oxf).* 2021;43:148–154.
- Pizot C, Boniol M, Mullie P, et al. Physical activity, hormone replacement therapy and breast cancer risk: a meta-analysis of prospective studies. *Eur J Cancer.* 2016;52:138–154.
- Monninkhof EM, Elias SG, Vlems FA, et al. Physical activity and breast cancer: a systematic review. *Epidemiology.* 2007;18:137–157.
- McTiernan A, Kooperberg C, White E, et al. Recreational physical activity and the risk of breast cancer in postmenopausal women: the Women's Health Initiative Cohort Study. *Jama.* 2003;290:1331–1336.
- Timmins IR, Jones ME, O'Brien KM, et al. International Pooled Analysis of Leisure-Time Physical Activity and Premenopausal Breast Cancer in Women From 19 Cohorts. *J Clin Oncol.* 2024;42:927–939.

Supplemental material

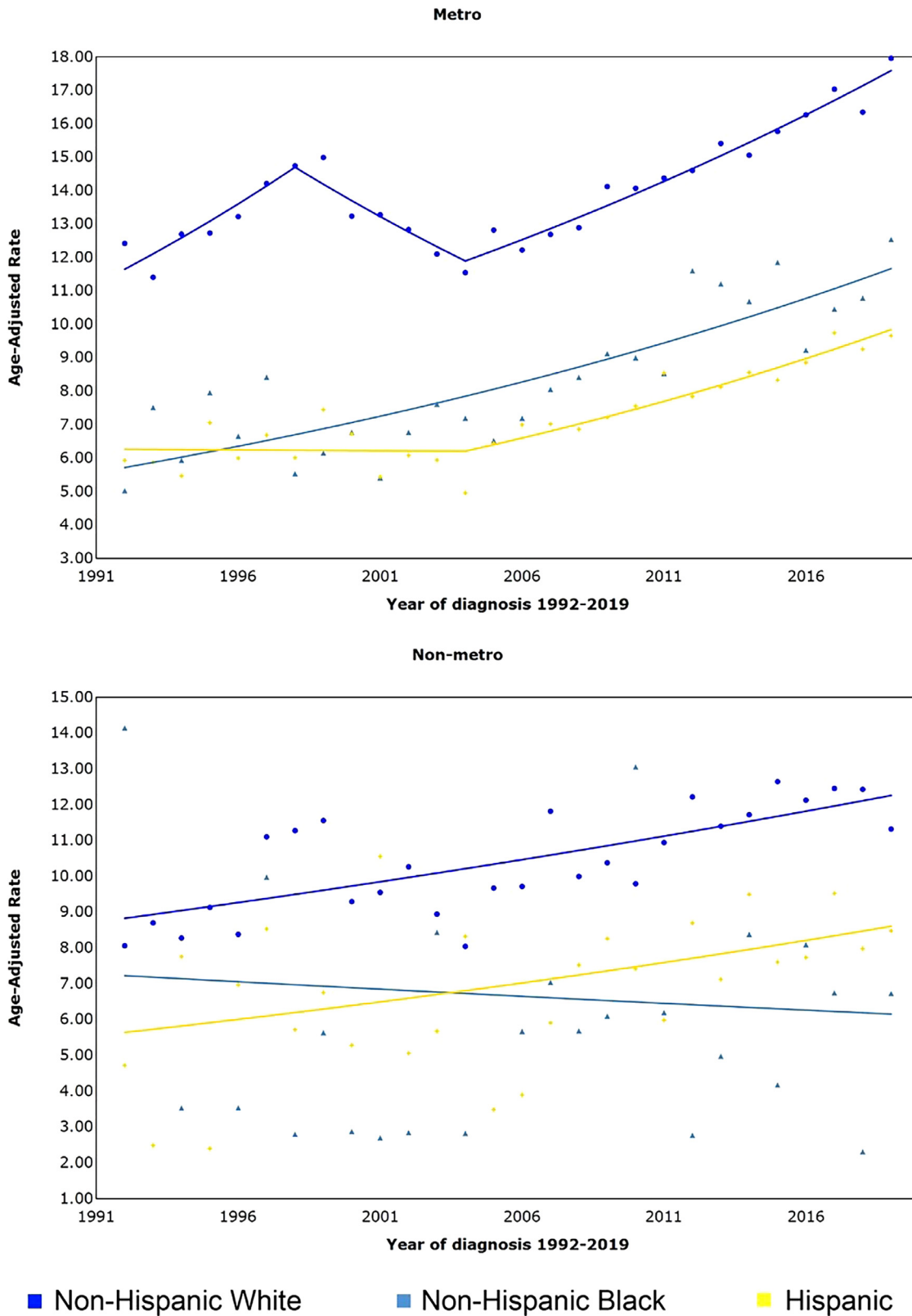
Supplemental Figure S1 Age-adjusted incidence rates of invasive lobular breast cancer by race/ethnicity stratified by stage from Surveillance, Epidemiology, and End Results Program (SEER) 12 Registries, 1992-2019. Age-adjusted incidence rate per 100,000 individuals.



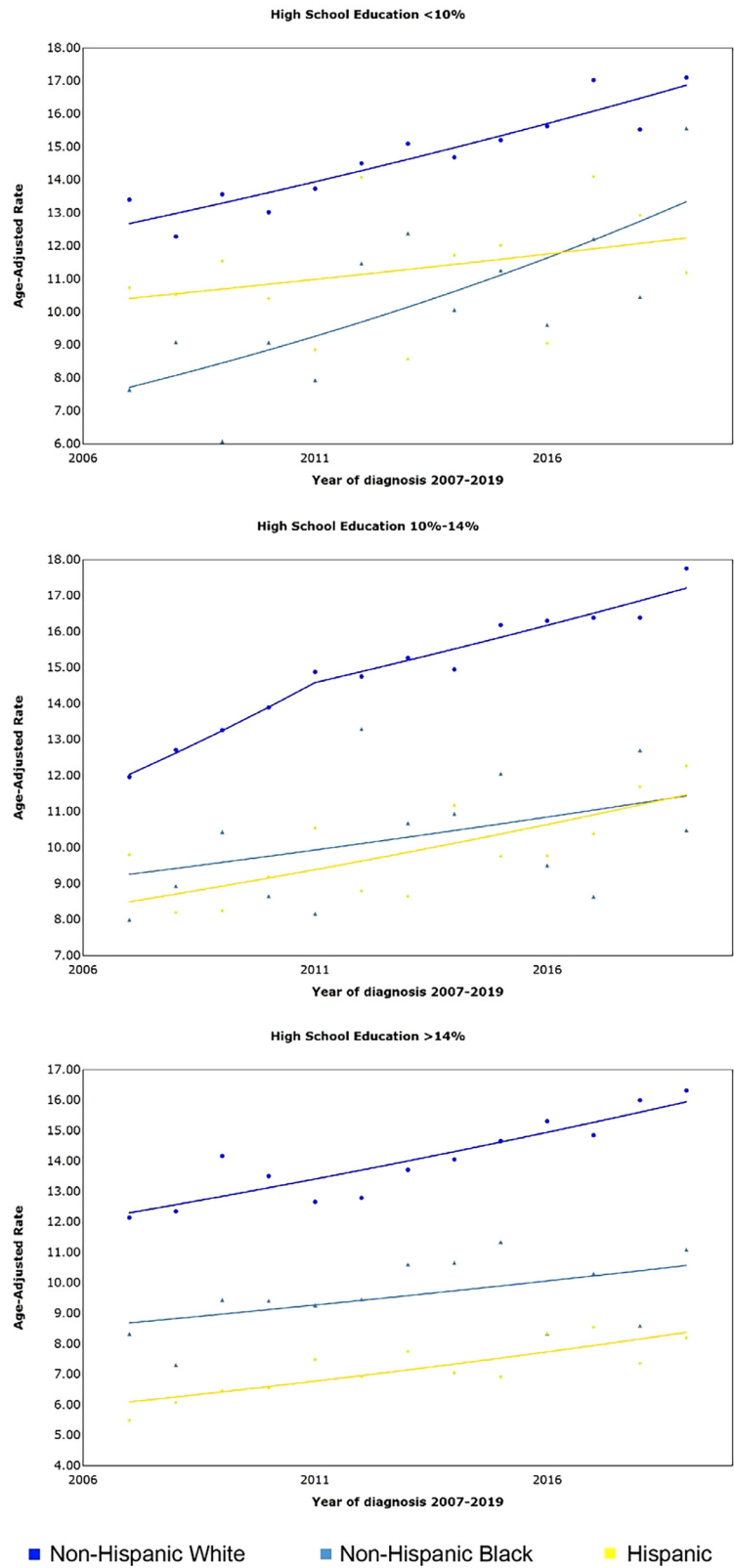
Supplemental Figure S2 Age-adjusted incidence rates of invasive lobular breast cancer by race/ethnicity stratified by poverty from Surveillance, Epidemiology, and End Results Program (SEER) 12 Registries, 1992-2019. Trends from level poverty were available only from 2007 to 2019. Age-adjusted incidence rate per 100,000 individuals.



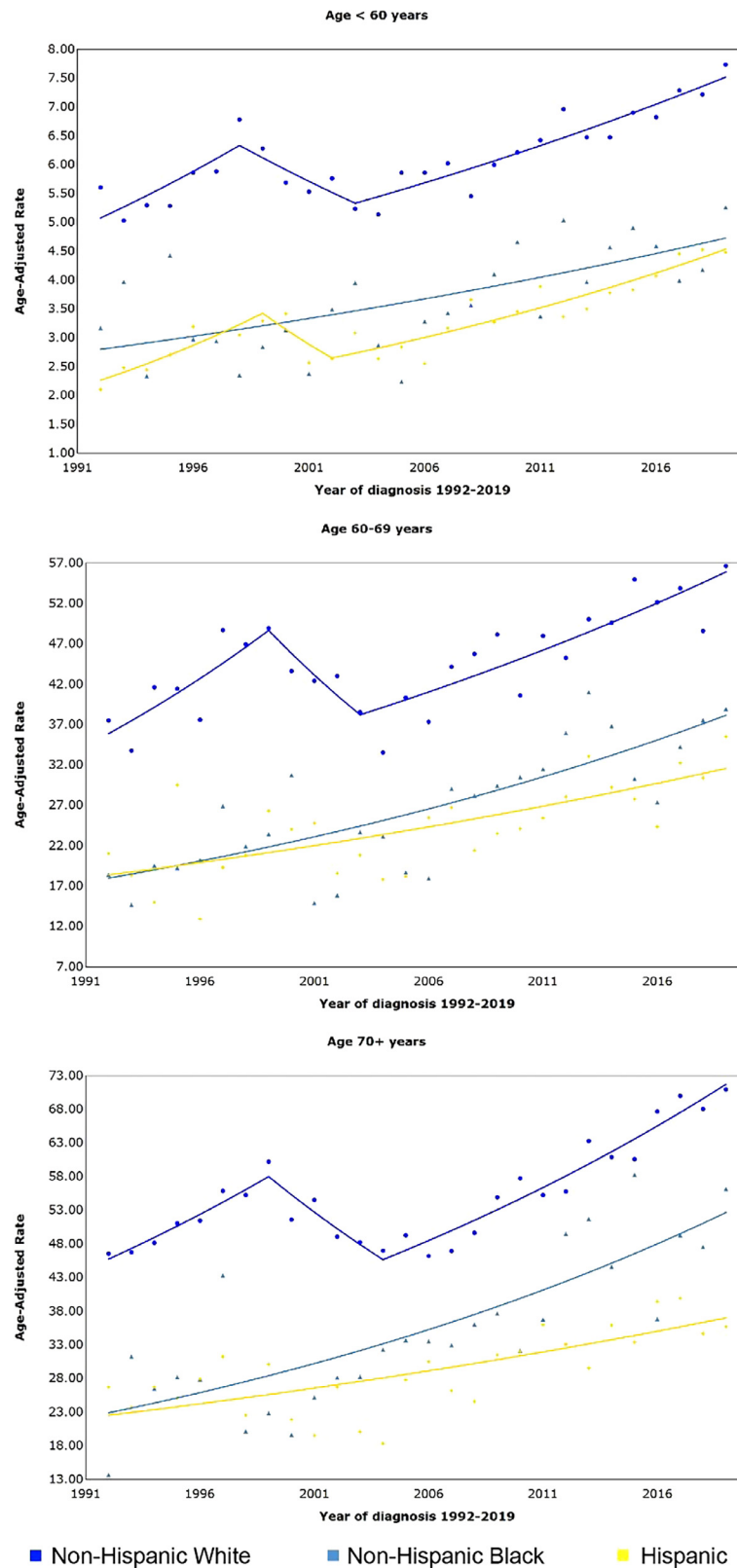
Supplemental Figure S3 Age-adjusted incidence rates of invasive lobular breast cancer by race/ethnicity stratified by county-level metro/nonmetro status from Surveillance, Epidemiology, and End Results Program (SEER) 12 Registries, 1992-2019. Age-adjusted incidence rate per 100,000 individuals.



Supplemental Figure S4 Age-adjusted incidence rates of invasive lobular breast cancer by race/ethnicity stratified by percentage of high school education from Surveillance, Epidemiology, and End Results Program (SEER) 12 Registries, 1992-2019. Age-adjusted incidence rate per 100,000 individuals.



Supplemental Figure S5 Age-adjusted incidence rates of invasive lobular breast cancer by race/ethnicity stratified by age groups from Surveillance, Epidemiology, and End Results Program (SEER) 12 Registries, 1992-2019. Age-adjusted incidence rate per 100,000 individuals.



Trends in Incidence of Invasive Lobular Carcinoma

Supplemental Table S1 Trends in Invasive Lobular Breast Cancer Incidence Rates by Race/Ethnicity and Stage Stratified by Socioeconomic Status and Education from Surveillance, Epidemiology, and End Results Program (SEER) 12 Registries, 1992-2019.

							APC			
			Trend 1		Trend 2		Trend 3		AAPC	
Stage	Strata	Race/Ethnicity	Years	APC	Years	APC	Years	APC	1992-2019	2007-2019
County-level % living in poverty ^a										
Local	< 10%	Non Hispanic White	2007-2019	+3.8*						+3.8*
		Non Hispanic Black	2007-2019	4.9						4.9
		Hispanic	2007-2019	+3.8*						+3.8*
	10%–19.9%	Non Hispanic White	2007-2019	+3.2*						+3.2*
		Non Hispanic Black	2007-2019	+2.8*						+2.8*
		Hispanic	2007-2019	+2.3*						+2.3*
	≥20%	Non Hispanic White	2007-2019	3.6						3.6
		Non Hispanic Black	2007-2019	−5.5						−5.5
		Hispanic	2007-2019	3.5						3.5
Regional	< 10%	Non Hispanic White	2007-2019	−0.5						−0.5
		Non Hispanic Black	2007-2019	2.6						2.6
		Hispanic	2007-2019	+3.7*						+3.7*
	10%–19.9%	Non Hispanic White	2007-2019	+0.9*						+0.9*
		Non Hispanic Black	2007-2019	0.8						0.8
		Hispanic	2007-2019	1.5						1.5
	≥20%	Non Hispanic White	2007-2019	2.5						2.5
		Non Hispanic Black	2007-2019	4.5						4.5
		Hispanic	2007-2019	3.0						3.0
Distant	< 10%	Non Hispanic White	2007-2019	0.7						0.7
		Non Hispanic Black	2007-2019	−2.4						−2.4
		Hispanic	2007-2019	2.7						2.7
	10%–19.9%	Non Hispanic White	2007-2019	0.6						0.6
		Non Hispanic Black	2007-2019	2.7						2.7
		Hispanic	2007-2019	1.8						1.8
	≥20%	Non Hispanic White	2007-2019	4.2						4.2
		Non Hispanic Black	2007-2019	−2.6						−2.6
		Hispanic	2007-2019	−4.8						−4.8
County-level metropolitan status										
Local	Metro	Non Hispanic White	1992-1998	+3.9*	1998-2004	−4.6*	2004-2019	+3.7*	+1.8*	
		Non Hispanic Black	1992-2002	−2.1	2002-2019	+4.6*			+2.1*	
		Hispanic	1992-2002	−1.4	2002-2019	+3.7*			+1.7*	
	Nonmetro	Non Hispanic White	1992-1999	+5.8*	1999-2003	−8.4	2003-2019	+3.0*	2.0	
		Non Hispanic Black	1992-2019	−0.8					−0.8	
		Hispanic	1992-2019	1.3					1.3	
Regional	Metro	Non Hispanic White	1992-1999	+4.9*	1999-2002	−5.6	2002-2019	+0.7*	1.0	
		Non Hispanic Black	1992-2019	+2.2*					+2.2*	
		Hispanic	1992-2019	+1.5*					+1.5*	

(continued on next page)

Supplemental Table S1 (continued)

							APC			
			Trend 1		Trend 2		Trend 3		AAPC	
Stage	Strata	Race/Ethnicity	Years	APC	Years	APC	Years	APC	1992-2019	2007-2019
	Nonmetro	Non Hispanic White	1992-2019	+1.2*					+1.2*	
		Non Hispanic Black	1992-2019	−1.5					−1.5	
		Hispanic	1992-2019	1.7					1.7	
Distant	Metro	Non Hispanic White	1992-2019	+1.9*					+1.9*	
		Non Hispanic Black	1992-2019	+2.4*					+2.4*	
		Hispanic	1992-2019	+1.9*					+1.9*	
	Nonmetro	Non Hispanic White	1992-2019	+2.2*					+2.2*	
		Non Hispanic Black	1992-2019	−2.2					−2.2	
		Hispanic	1992-2019	−1.1					−1.1	
% Highschool education ^a										
Local	< 10%	Non Hispanic White	2007-2019	+3.6*						+3.6**
		Non Hispanic Black	2007-2019	+5.3*						+5.3*
		Hispanic	2007-2019	1.4						1.4
	10-14%	Non Hispanic White	2007-2019	+4.0*						+4.0*
		Non Hispanic Black	2007-2019	2.7						2.7
		Hispanic	2007-2019	+2.5*						+2.5*
	> 14%	Non Hispanic White	2007-2019	+3.3*						+3.3*
		Non Hispanic Black	2007-2019	1.9						1.9
		Hispanic	2007-2019	+3.1*						+3.1*
Regional	< 10%	Non Hispanic White	2007-2019	0.4						0.4
		Non Hispanic Black	2007-2019	1.9						1.9
		Hispanic	2007-2019	1.6						1.6
	10-14%	Non Hispanic White	2007-2019	0.7						0.7
		Non Hispanic Black	2007-2019	1.0						1.0
		Hispanic	2007-2019	2.3						2.3
	> 14%	Non Hispanic White	2007-2013	−2.7	2013-2019	3.5				0.3
		Non Hispanic Black	2007-2019	0.3						0.3
		Hispanic	2007-2019	+2.0*						+2.0*
Distant	< 10%	Non Hispanic White	2007-2019	0.6						0.6
		Non Hispanic Black	2007-2019	2.7						2.7
		Hispanic	2007-2019	−5.3						−5.3
	10-14%	Non Hispanic White	2007-2019	0.9						0.9
		Non Hispanic Black	2007-2019	−1.1						−1.1
		Hispanic	2007-2019	5.4						5.4
	> 14%	Non Hispanic White	2007-2019	1.4						1.4
		Non Hispanic Black	2007-2019	5.1						5.1
		Hispanic	2007-2019	1.5						1.5
Age groups										
Local	< 60 years	Non Hispanic White	1992-1998	+3.8*	1998-2003	−5.2	2003-2019	+3.4*	+1.9*	
		Non Hispanic Black	1992-2003	−2.7	2003-2019	+4.4*			1.5	
		Hispanic	1992-2005	−0.2	2005-2019	+4.5*			+2.2*	
	60-69 years	Non Hispanic White	1992-1997	5.9	1997-2004	−3.6	2004-2019	+3.5*	+2.0*	

(continued on next page)

Trends in Incidence of Invasive Lobular Carcinoma

Supplemental Table S1 (continued)

Stage	Strata	Race/Ethnicity	APC							
			Trend 1		Trend 2		Trend 3		AAPC	
			Years	APC	Years	APC	Years	APC	1992-2019	2007-2019
Regional	≥ 70 years	Non Hispanic Black	1992-2019	+2.8*					+2.8*	
		Hispanic	1992-2019	+2.1*					+2.1*	
		Non Hispanic White	1992-1999	+3.7*	1999-2004	−6.2*	2004-2019	+3.8*	+1.9**	
		Non Hispanic Black	1992-2000	−5.3	2000-2019	+5.3*			2.0	
		Hispanic	1992-2019	+2.1*					+2.1*	
	< 60 years	Non Hispanic White	1992-2019	+0.7*					+0.7*	
		Non Hispanic Black	1992-2019	+1.6*					+1.6*	
		Hispanic	1992-1995	27.2	1995-2019	+0.9*			3.6	
		Non Hispanic White	1992-1998	5.7	1998-2019	−0.3			1.0	
		Non Hispanic Black	1992-2019	+2.5*					+2.5*	
	60-69 years	Hispanic	1992-2019	+1.9*					+1.9*	
		Non Hispanic White	1992-2019	+0.6*					+0.6**	
		Non Hispanic Black	1992-2019	+2.4*					+2.4*	
		Hispanic	1992-2019	+1.5*					+1.5*	
		Non Hispanic White	1992-2019	0.5					0.5	
Distant	< 60 years	Non Hispanic Black	1992-2019	2.0					2.0	
		Hispanic	1992-2019	+2.1*					+2.1*	
		Non Hispanic White	1992-2019	+1.8*					+1.8*	
		Non Hispanic Black	1992-2019	+3.3*					+3.3*	
		Hispanic	1992-2005	−9.6*	2005-2009	28.4	2009-2019	−1.4	−1.7	
	60-69 years	Non Hispanic White	1992-2019	+3.2*					+3.2*	
		Non Hispanic Black	1992-2019	1.9					1.9	
		Hispanic	1992-2019	0.6					0.6	
	≥ 70 years	Non Hispanic White	1992-2019	+3.2*					+3.2*	
		Non Hispanic Black	1992-2019	1.9					1.9	
		Hispanic	1992-2019	0.6					0.6	

Abbreviations: APC = annual percent change; AAPC = average annual percent change.

*Statistically significant ($P < .05$).

^a Trends for county-level poverty, household income, and high school education were available only from 2007 to 2019.